ARTHUR RAMANDALINA



CIVIL & STRUCTURAL ENGINEER | PHD

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PROFESSIONAL PROFILE

Civil and structural engineer with more than ten years of experience across building structures, public infrastructure, engineering coordination and field operations. Prepared structural analyses and design documentation for projects serving French clients; contributed to road, bridge and disaster-recovery planning in Madagascar; and now develop customer-driven Fix3RD products in Japan. Completed a PhD in seismic isolation research, combining analytical methods, experimental evidence and practical construction considerations.

ENGINEERING STRENGTHS

Structural design & analysis • Roads and bridges • Product development for manufacture • Technical drawing & quantity estimation • Site investigation • Experimental mechanics & seismic isolation

PROFESSIONAL EXPERIENCE

Fix3RD Engineer | STV 2025–Present

Chiba, Japan

- Develop Fix3RD products jointly developed by Ooparts and STV, converting customer and site requirements into engineering concepts, detailed CAD documentation and specifications for manufacturing handoff.
- Resolve design questions directly with customers, partner companies and suppliers; incorporate practical feedback, handling conditions and fabrication constraints into design revisions.
- Prepare construction-related drawings and quantity estimates and use site visits to inform design decisions and check relevant civil-engineering works.

Doctoral Researcher, Structural Dynamics Laboratory | Saitama University Apr 2022–Sep 2025

Saitama, Japan

- Planned and conducted experimental research on laminated silicone elastomeric bearings for seismic isolation, from specimen design and fabrication to mechanical testing and interpretation.
- Assessed tensile and cyclic compression–shear responses, equivalent stiffness and damping; compared findings with relevant test standards and prepared research presentations and manuscripts.

Project Designer / Technical Engineer | Ministry of Public Works Sep 2019–Mar 2022

Madagascar

- Supported national road and bridge programmes through site investigations, preliminary engineering studies, design documentation, cost estimates and procurement materials.
- Contributed to post-disaster infrastructure assessments and recovery planning, including technical reports, proposed interventions and implementation schedules.
- Worked on RN7 bridge proposals, RN1–RN4 interchange planning and an NR26 hydropower-access road scheme; specific contributions are summarized on page 2.

Chief Road Control Station Officer | Road Administration Agency Nov 2013–Jun 2019

Madagascar

- Directed daily operation of a national road control station, coordinating personnel, logistics, road-user compliance activities and traffic/weight-control data reporting.
- Supported maintenance planning and enforcement coordination; contributed to station layouts, operating procedures and digital reporting concepts for axle-load control.

EARLIER ENGINEERING EXPERIENCE

Structural Designer | Netco Bat Jan 2012–Oct 2013

Madagascar · Projects for clients in France

- Performed structural calculations and developed technical and construction drawings for commercial, institutional and healthcare buildings, including refurbishment projects.
- Worked on structural arrangements, load checks, temporary works considerations and coordination of design details with project teams.

Site Engineer | Enterprise Frederic Jan 2010–Dec 2011

Madagascar

- Supervised reinforced-concrete activities, checked reinforcement and formwork against drawings, prepared quantity take-offs and maintained site progress records.

Additional engagement: Cadastral and land-use planning consultant, Ministry of Spatial Planning, Madagascar (2011–2013; overlapping appointments). Produced subdivision layouts and technical documentation in coordination with surveying and administrative stakeholders.

SELECTED PROJECT EVIDENCE

RN7 bridge proposals | Madagascar, 2021. Contributed to site investigations, preliminary design and working drawings, and budget/schedule inputs for two proposed bridges (60 m and 80 m spans).

NR26 hydropower access-road scheme | Madagascar, 2020. For an 85 km road-upgrade programme, prepared alignment alternatives, preliminary cost inputs and technical correspondence with hydropower planners.

RN1–RN4 interchange concept | Madagascar. Assisted with geometric layout, traffic-related studies, budget planning and tender documentation for a proposed multi-level interchange.

French healthcare and municipal buildings | 2012–2013. Prepared calculations, load checks and working drawings for healthcare buildings and institutional renovations, including laboratory-equipment loading and structural refurbishment details.

EDUCATION

PhD in Civil Engineering | Saitama University, Japan | 2025. Research: experimental study of silicone elastomeric bearings for seismic isolation (Structural Dynamics Laboratory).

Engineering Diploma — Building and Public Works | École Supérieure Polytechnique d'Antananarivo, University of Antananarivo, Madagascar | 2010. Specialization: Building and Public Works.

ADDITIONAL JAPANESE INDUSTRY EXPOSURE

Takigami Steel Construction · Short industrial internship · September 2025. Practical exposure to steel fabrication and welding quality control, bridge construction and maintenance-site operations, and a supervised DIANA FEA exercise.

TECHNICAL TOOLS & LANGUAGES

Engineering software. AutoCAD (2D/3D), Revit, ArchiCAD, Jw_cad, Autodesk Robot Structural Analysis, DIANA FEA, Abaqus; familiarity with other finite-element analysis tools.

Computation & reporting. MATLAB, Python, VB; engineering calculations, experimental data processing and Microsoft Office.

Languages. English — fluent; French — fluent; Japanese — conversational; continuing professional language study.